

Spoken Tutorial – Code and Audio Generation (The Specialists)

Title: Code and Audio Generation (The Specialists)

Module	Generative AI Basics
Episode	4.3
Learning Objective	At the end of this tutorial learner will be able to 1. Understand how Generative AI assists in creating code and audio.
Approx. Duration	5 minutes
Outline	• Introduction to Code and Audio Generation • Code Generation: GitHub Copilot as a developer's assistant • How AI helps in writing code and finding bugs • Audio Generation: Voice synthesis and text-to-speech • Music creation using AI models
Meta Tags	Generative AI, Code Generation, Audio Generation, GitHub Copilot, Voice Synthesis, Text-to-Speech, AI Music, Spoken Tutorial
Pre-requisite Tutorial	Basic computer knowledge and an understanding of what Generative AI is.

Script

Visual Cue	Narration
A futuristic digital landscape with code snippets and musical notes floating around, representing AI's ability to create both.	Welcome to this Spoken Tutorial on Code and Audio Generation. In this lesson, we will see how Generative AI helps make new code and audio.
A clear, focused image of a light bulb turning on, symbolizing understanding or a learning objective.	After this tutorial, you will know how Generative AI helps to make code and audio.
A simple drawing of a computer monitor with a Wi-Fi symbol, indicating internet connectivity.	To follow this tutorial, you need a computer with internet. You do not need to install any special software.
A stack of books labeled 'Basic Computer Knowledge' and 'Generative AI Intro', showing foundational learning.	Before you start, you should know basic computer things. Also, you should know what Generative AI is and what it can do.
An abstract illustration showing different forms of media (text, image, code, audio waves) coming from a central AI brain.	Generative AI is very strong. It can make new text and pictures. Now, it can also make new code and audio.
A stylized image of a robot hand typing on a keyboard, with code lines appearing on a screen, representing AI coding assistance.	Let us start with making code. Tools like GitHub Copilot are good examples. These tools learn from many existing codes to help you.

A programmer working at a desk with a glowing AI assistant icon next to them, showing collaboration.	These tools do not take the place of human programmers. They work like a very smart helper. They help programmers write code faster and better.
A screen showing a code editor with AI-generated suggestions popping up as the user types.	How does it help? When you write code, Copilot can suggest the next lines of code. This saves much time for programmers.
A split screen showing a comment being translated into code on one side and a magnifying glass highlighting a bug in code on the other.	It can turn your comments into code. For example, if you write 'add two numbers', it can write the code. It also helps find mistakes, or bugs, in your code.
Sound waves transforming into musical notes and speech bubbles, with an AI brain icon in the background.	Now, let us look at making audio. These AI tools learn from many sounds, voices, and music. Then they can make new audio.
A microphone icon with sound waves coming out, and text lines transforming into spoken words.	One thing AI can do is make voices. This means making sounds like a real human voice. Text-to-Speech tools use this to read out written words.
Icons representing an audiobook, a podcast microphone, and a virtual assistant (like a smart speaker).	You can use these AI voices for many things. For example, they can read audiobooks. They can also make voices for podcasts or virtual helpers.
Musical notes flowing from an AI icon, forming a beautiful melody. A video camera icon is subtly present.	AI can also make music. It can create background music for your videos. You can use this music freely, it is called royalty-free music.
A diverse collection of musical instruments and a conductor's baton, with AI generating various music styles.	AI can also make new tunes or melodies. It can even make songs like a famous artist's style. This helps create music in new ways.
A summary infographic showing two main branches: 'Code' with a keyboard icon, and 'Audio' with a speaker icon, both originating from an AI brain.	In this tutorial, we learned about Generative AI in two areas. We saw how it helps make code with tools like GitHub Copilot. We also saw how it makes audio, like voices and music.
A thought bubble with a light bulb and question mark, suggesting an activity or exploration.	For an assignment, search for an online text-to-speech tool. Type a sentence and hear an AI voice. Or, find some royalty-free music made by AI.
The logo of EduPyramids Educational Services Private Limited and IIT Bombay SINE, with a 'Thank You' message.	This Spoken Tutorial is from EduPyramids Educational Services Private Limited, SINE, IIT Bombay. Thank you for joining us!